Primer: Adversarial Inverse Reinforcement Learning (AIRL)

econirl package documentation

1 Problem and Motivation

Standard IRL methods recover a reward function from observed behavior, but the recovered reward is not robust to changes in the environment. For any function $h : \mathcal{S} \to \mathbb{R}$, the shaped reward $r'(s,a) = r(s,a) + \gamma h(s') - h(s)$ induces the same optimal policy under the original dynamics but a different policy under new dynamics. AIRL (Fu et al. 2018) solves this by decomposing the discriminator into a transferable reward $g_\theta(s,a)$ and a non-transferable shaping potential $h_\phi(s)$, learning them jointly so that g_θ converges to the true reward. Lee, Sudhir, and Wang (2026) extend AIRL with anchor constraints and latent heterogeneity for structural estimation of dynamic discrete choice models on serialized content platforms. We call this Anchored AIRL (AAIRL).

2 Model and Notation

The agent solves the maximum causal entropy problem

$$\max_{\pi} \; \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t) + \sigma \mathcal{H}(\pi(\cdot \mid s_t)) \big) \right] \tag{1}$$

where $\mathcal{H}$ is the Shannon entropy. The optimal policy satisfies

$$\pi^*(a \mid s) = \frac{\exp(A^*(s,a)/\sigma)}{\sum_{a'} \exp(A^*(s,a')/\sigma)}, \qquad A^*(s,a) = Q^*(s,a) - V^*(s). \tag{2}$$

Symbol	Meaning
$s \in \mathcal{S}$, $a \in \mathcal{A}$, s'	State, action, next state
γ, σ	Discount factor, entropy scale ($\sigma = 1$ in IRL convention)
$r^*(s,a)$, $\pi^*(a \mid s)$	True reward, expert policy
$P(s' \mid s,a)$	Transition kernel (known for tabular, unknown for neural)
$V^*(s)$, $Q^*(s,a)$, $A^*(s,a)$	Value, action-value, advantage functions
$\tau = (s_0, a_0, s_1, \ldots)$	Trajectory
$g_\theta(s,a)$	Reward network (parameterized by θ)
$h_\phi(s)$	Shaping potential $\approx V(s)$ (parameterized by ϕ)
$\pi_\omega(a \mid s)$	Policy network (parameterized by ω)
$f_{\theta,\phi}(s,a,s')$	Discriminator logit (Eq. 3)
$D_{\theta,\phi}(s,a,s')$	Discriminator output $\in [0,1]$ (Eq. 4)

AIRL uses a structured discriminator that decomposes into reward and shaping:

$$f_{\theta,\phi}(s,a,s') = g_\theta(s,a) + \gamma\, h_\phi(s') - h_\phi(s), \tag{3}$$

$$D_{\theta,\phi}(s,a,s') = \frac{\exp(f_{\theta,\phi})}{\exp(f_{\theta,\phi}) + \pi_\omega(a \mid s)}. \tag{4}$$

The key structural assumption is in (3): the logit f is a sum of a state-action reward term and a potential-based shaping term, mirroring the Bellman decomposition $Q = r + \gamma\mathbb{E}[V(s')] - V(s) + V(s)$.

3 Identification

Reward shaping ambiguity. IRL rewards are identified only up to potential-based shaping (Ng et al. 1999). The observed policy constrains $|\mathcal{S}|(|\mathcal{A}| - 1)$ equations but the reward has $|\mathcal{S}| \cdot |\mathcal{A}|$ unknowns, leaving $|\mathcal{S}|$ free variables corresponding to the value function (Cao and Cohen 2021, Theorem 1).

AIRL disentanglement (Fu et al. 2018, Theorem 1). At the adversarial optimum, if the reward is state-only (g_θ depends on s not a) and the MDP is decomposable (all states linked), then $g_\theta(s) = r^*(s) + c$ for some constant c. The decomposability condition rules out absorbing states, limiting applicability to DDC models.

Anchor identification (Lee, Sudhir, Wang 2026). Anchoring extends the result to action-dependent rewards. If one action (exit) has known zero reward and one state (absorbing) has known zero value,

$$g_\theta(s, a_{\text{exit}}) = 0 \;\; \forall s, \qquad h_\phi(s_{\text{abs}}) = 0, \tag{5}$$

the decomposition (3) is unique at convergence: $g_\theta(s, a) = r^*(s, a)$ and $h_\phi(s) = V^*(s)$.

4 Estimation Algorithm

Algorithm 1 AIRL with Anchor Constraints

Require: Expert transitions $\mathcal{D}_{\text{obs}} = \{(s, a, s')_n\}_{n=1}^N$, discount γ, learning rate α, anchor action a_{exit}, absorbing state s_{abs}

Ensure: Recovered reward g_θ, value function h_ϕ, policy π_ω

```
Initialize θ_0, φ_0, ω_0; set π_ω ← uniform
for round = 1, 2, ... until convergence do
    // Generate policy transitions
    Sample M transitions D_sim by rolling out π_ω
    // Update discriminator
    for j = 1, ..., J do
        Compute f_n via (3), then D_n via (4)
        _disc = −(1/N) Σ_{D_obs} log D_n − (1/M) Σ_{D_sim} log(1 − D_m)
        θ, φ ← θ, φ − α ∇_{θ,φ} disc
        Enforce anchors (5)
    end for
    // Update policy
    Reward signal: r̃ = f_{θ,φ}(s, a, s') − log π_ω(a | s)
    Tabular: π_ω ← (1 − α_p) π_ω^old + α_p π^VI    (conservative mixing)
    Neural: update ω via PPO on max_{π_ω} E[Σ_t γ^t r̃]
    if max_{s,a} |π^new − π^old| < ε then
        break
    end if
end for
```

Generator options. The tabular generator uses soft value iteration with conservative policy mixing ($\alpha_p < 1$) to prevent reward divergence from exact VI overshooting. The neural generator uses PPO with clipped surrogate objective, which naturally takes small policy steps.

At convergence. When $D = 1/2$ everywhere: $f_{\theta,\phi} = \log \pi^* = A^*$. With anchors (5) enforced, $g_\theta = r^*$ and $h_\phi = V^*$.

5 Simulation Example

We test AIRL on the serialized content environment from `examples/serialized-content/lsw_replication.py`. The environment has 51 states (50 episodes across 5 books plus one absorbing state), 3 actions (buy, wait, exit), and 6 linear reward features. The ground truth is Segment A ("Pay and Read") from the LSW replication. Data consists of 2,000 users reading 3 books each, producing 5,997 trajectories and 67,391 observations.

Parameter	True	NFXP	AIRL (α_p=0.1, 500 rds)
buy_cost	−0.30	−0.06	4.90
quality_buy	2.00	1.72	4.90
cliffhanger_buy	1.50	1.54	4.92
wait_cost	−0.80	−0.79	1.79
quality_wait	0.50	0.49	2.41
progress_buy	0.80	0.72	4.86

Table 1: Parameter recovery on 51-state serialized content (K=1). Run `python airl_run.py` to regenerate. NFXP recovers all parameters with correct signs. AIRL with conservative VI still shows reward scale inflation, though relative feature ordering is preserved.

NFXP is the gold standard on this tabular problem: it solves the Bellman fixed point exactly and recovers all 6 parameters within 15% of the true values in 13 seconds. AIRL with conservative policy iteration slows the reward divergence (reward range 5 vs 50 without dampening) but does not fully stabilize. The relative ordering of buy-action features (quality > cliffhanger > progress > buy cost) is preserved, but the absolute values are inflated by a factor of 3–5. This is the known adversarial reward scale issue: the discriminator loss is invariant to uniform scaling of the reward, so without an external anchor on the scale, the optimizer drifts upward. The anchor constraints (5) pin down the level but not the scale.

The three counterfactual exercises (Type A: introduce WFF, Type B: customized wait-times, Type C: content-based pricing) use ground truth rewards and produce qualitatively correct results. Wait and Read purchases increase 356% under content-based pricing. These counterfactuals do not depend on the AIRL recovery quality because they use the known structural parameters.

6 Comparison to Other Estimators

	NFXP	MCE-IRL	AIRL	GLADIUS
Requires known P	Yes	Yes	No (PPO) / Yes (VI)	Yes
Continuous states	No	No	Yes (PPO)	Yes
Action-dependent r	Yes	Yes	With anchors	Yes
Identification	Parametric MLE	Feature matching	Disentanglement	Bellman decomp.
Tabular performance	Excellent	Good	Unstable	Biased

Table 2: AIRL in context. AIRL's advantage is continuous-state capability via PPO. On tabular problems, structural methods (NFXP, MCE-IRL) are more stable and accurate.

AIRL's advantage over NFXP and MCE-IRL is handling continuous state spaces when using PPO as the generator. Its advantage over GLADIUS is the disentanglement guarantee: the recovered reward is structurally meaningful, not a Q-value identified only up to a state-dependent constant. On tabular problems, NFXP dominates: faster, more accurate, and theoretically grounded in maximum likelihood.

References.

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